1 Algebraic Extensions and Homomorphisms

Formally, field extensions L/K are homomorphisms $K \to L$, but in most cases, we assume the homomorphism is inclusion anyway. From here on we always write the minimal polynomial (over K) of $\alpha \in L$, where L/K is an extension, is denoted as $m_K(\alpha)$.

Theorem 1. *If $L = K(\beta)$, $\beta \in L$, then $[L : K] = \deg m_K(\beta)$.*

In particular, $\beta \in L$ is transcendental over K, if and only if one have $[K(\beta) : K] = \infty$,

Theorem 2. *Let $L : K$ be a field extension, and assume $K \subseteq L$. The following are equivalent:*

1. *$[L : K] < \infty$.*
2. *$L : K$ is an algebraic extension, **and** there are $\alpha_1, \ldots, \alpha_n \in L$ so that $L = K(\alpha_1, \ldots, \alpha_n)$ (where $K(\alpha_1, \ldots, \alpha_n)$ denotes the smallest subfield of L containing K and $\alpha_1, \ldots, \alpha_n$).*
3. *There are $\alpha_1, \ldots, \alpha_n \in L$ so that $\alpha_1, \ldots, \alpha_n$ are algebraic over K and $L = K(\alpha_1, \ldots, \alpha_n)$.*

Of course, not all algebraic extensions are finite, but all finite algebraic extensions look like this.

The following theorems concern about "liftings". I skipped a few setups, but I hope the message is clear:

Theorem 3. *If $\sigma : K_1 \to K_2$ is a field **isomorphism**, then it extends to $\tau : K_1(\alpha) \to K_2(\beta)$ **isomorphism** if and only if $\sigma(m_{K_1}(\alpha)) = m_{K_2}(\beta)$.*

Theorem 4. *Let $L/M/K$ be an algebraic tower, For any K-homomorphism $\sigma : M \to \overline{K}$, there is always an extension to a K-homomorphism $\tau : L \to \overline{K}$ such that $\tau|_M = \sigma$.*

Definition 1. Define $\mathrm{Gal}(L/K) = \{\sigma : \sigma : L \to L \text{ is a } K\text{-isomorphism }\}$.

2 Separable and Normal Extensions

Definition 2. A polynomial $f \in K[x]$ **splits** over L if f has linear factors when written over L.

Definition 3. A polynomial $f \in K[x]$ is **separable over** K if all the irreducible components don't have repeated roots (no "> 1 powers" in the factorization in $\overline{K}$).

Definition 4. Given an extension L/K, an element $\alpha \in L$ is **separable over** K if $m_K(\alpha) \in K[x]$ is separable over K. If all such αs are separable, L is called **separable over** K.

Theorem 5 (Derivative Test)**.** *Take $f \in K[t] \backslash K$ not necessarily irreducible, and let $L : K$ be a field extension so that f splits over L. Assume that $K \subseteq L$. The following are equivalent:*

- *The polynomial f has a repeated root in L.*
- *There is some $\alpha \in L$ so that $f(\alpha) = 0 = f'(\alpha) \Leftrightarrow \gcd(f, f') \neq$ constant.*

Theorem 6 (Separability Check - Polynomial)**.** *If $f \in K[x]$ is irreducible:*

- *If $\mathrm{Char}(K) = 0$, then every f is separable over K.*
- *If $\mathrm{Char}(K) = p$, a prime, then f is inseparable $\Leftrightarrow f' \equiv 0 \Leftrightarrow f$ is in the form of $g(x^p)$ for some $g \in K[x]$.*

All elements in $\mathbb{Q}[x], \mathbb{F}_q[x]$, where q is prime powers, is always separable. So any field extension base on it is always separable.
An inseparable polynomial example is $x^p - t$ in the extension $\mathbb{F}_p(t)(\sqrt[p]{t})/\mathbb{F}_p(t)$.

Theorem 7 (Primitive Element Theorem)**.** *If $[L : K] < \infty$ and L/K separable, then there is $\alpha \in L$ such that $L = K(\alpha)$.*

Theorem 8 (Lifting Lemma, reprise)**.** *Given field homomorphism $\sigma : K \to \overline{K}$, there is always $\widehat{\sigma} : L \to \overline{K}$ such that $\widehat{\sigma}|_K = \sigma$.*

- *Moreover, if L/K is a finite extension, if L/K is separable, then there is exactly $[L : K] := n$ of such distinct $\widehat{\sigma}$. Otherwise, it has no more than n such $\widehat{\sigma}$.*

Theorem 9 (Equivalent Definition of Normacy)**.** *If L/K is a finite extension, the following are equivalent:*

1. *For all the irreducible polynomial $f \in K[x]$, if $\alpha \in L$ is as root of f, then all the roots of f (considered as a factorization in $\overline{K}$) are in L.*
2. *L is the splitting field for some set of polynomial S over K.*

*Both definition can be used for L/K being **normal**, note that normal extensions are always algebraic.*

Note the two theorem regarding transitivity:

Theorem 10 (Transitivity of Separability)**.** *If $L/M/K$ is tower of extension, L/K separable $\Leftrightarrow$ $L/M, M/K$ separable.*

Theorem 11 (Transitivity of Normacy)**.** *If $L/M/K$ is a tower, L/K normal, TFAE*

1. *M/K normal.*
2. *For any $\psi : M \to L$ a K-homomorphism, $\psi(M) = M$.*
3. *For $\sigma : L \to L$ is a K-isomorphism, we have $\sigma(M) \subseteq M$.*

To check a field is normal, the easiest way probably is by condition 3, and note that a splitting field of S over K can be realized as the smallest field that contains K, and contains the set $R = \{\alpha : \alpha \text{ is a root for some polynomial } f \in S\}$.

There is also a theorem of checking whether a splitting field is separable. This synergizes with the equivalent definition of seperability above.

Theorem 12. *If L is a splitting field of a set of polynomials S over K, then L/K is separable if and only if every polynomial $f \in S$ is separable over K.*

3 Some Galois Theory

Definition 5. A normal and separable extension is called **Galois**.

Key examples of splitting field over $\mathbb{Q}$ are $x^3 - 2$ (roots are $\sqrt[3]{2}$, $\sqrt[3]{2}\omega$, and $\sqrt[3]{2}\omega^2$, where ω is the cube root of unity) and $8x^3 + 4x^2 - 4x - 1$ (roots are $\cos\left(\frac{2k\pi}{7}\right), k = 1, 2, 3$). They have degrees of 6 and 3 respectively. See Section 6.3 on Ash for a detailed calculation, their Galois Group is S_3 and $\mathbb{Z}/3\mathbb{Z}$ respectively.

Definition 6. Let $G = \mathrm{Gal}(L/K)$ be the Galois group of the extension L/K. If H is a subgroup of G, the **fixed field** of H is

$$\phi(H) = \{x \in L : \sigma(x) = x \text{ for every } \sigma \in H\}.$$

If $K \leq M \leq L$, define the "**fixed group**" of M to be

$$\gamma(M) = \mathrm{Gal}(L/M) = \{\sigma \in G : \sigma(x) = x \text{ for every } x \in M\}$$

(we used quotes because the name is not standard.)

Theorem 13 (Dedekind)**.** *Distinct "characters" (for now, the homomorphism in Galois Group $\mathrm{Gal}(L/K)$) are linearly independent over L.*

Theorem 14 (Galois Correspondence)**.** *Let $K \leq M \leq L$ be a tower of fields, L/K Galois and finite, then:*

- *ϕ is a bijective map from subgroups to intermediate fields, with inverse γ.*
- *(Reverse Inclusion.) If $H_1 \leq H_2$ then $\phi(H_1) \geq \phi(H_2)$, and if $M_1 \leq M_2$, then $\gamma(M_1) \geq \gamma(M_2)$.*
- *Suppose that the intermediate field M corresponds to the subgroup H under the Galois correspondence. i.e. $\mathrm{Gal}(L/M) = H$, i.e. $\phi(H) = M$, Then*
  - *L/M is always a normal field extension (hence Galois);*
  - *$[M : K] = [G : H]$ and $[L : M] = |H|$*
  - *Kor any $\sigma \in \mathrm{Gal}(L/K)$, then the field $\sigma M = \{\sigma(x) : x \in M\}$ corresponds to the conjugate subgroup $\sigma H \sigma^{-1}$. i.e. $\mathrm{Gal}(L/\sigma M) \cong \sigma H \sigma^{-1}$.*
  - *M/K is a normal field extension $\Leftrightarrow$ H is a normal subgroup of G, and $\mathrm{Gal}(M/K) \cong G/H$.*

4 Special Types of Extensions

Definition 7.
- A **cyclotomic extension** L/K is a field extension where L is the splitting field of the polynomial $x^n - 1$.
- The roots of $x^n - 1$ in $\overline{K}$ are called the **roots of unity**, all the roots of unity form a $\leq n$-element, NOT NECESSARILY EQUAL cyclic subgroup of L^*.
- A **primitive** n^{th} **root of unity**, if it exists, is a generator of the above subgroup, which is of order n.

It is obvious that if a primitive n^{th} root of unity does exist, call it ω_n, then $L = K(\omega_n)$.

The non-existence of primitive roots can be demonstrated by the example of $x^{20} - 1 = (x^4 - 1)^5$ over $\mathbb{F}_5$.

Theorem 15. *If ω_n is a primitive root of unity over $\mathbb{Q}$, then $\mathrm{Gal}(\mathbb{Q}(\omega_n)/\mathbb{Q}) = U_n$, the multiplicative group of $\phi(n)$ elements.*

The proof of the theorem introduces the following powerful polynomial:

Definition 8. The nth **cyclotomic polynomial**, ψ_n, is

$$\psi_n(x) := \prod_i (x - \omega_i)$$

where $\omega_1, \omega_2 \cdots$ are n^{th} primitive roots of unity, so there is exactly $\phi(n)$ of them. (why, *a priori*?)

Theorem 16. *The cyclotomic polynomial is actually irreducible over $\mathbb{Q}$ and has integer coefficients. Moreover,*

$$x^n - 1 = \prod_{d|n} \psi_d(x).$$

With $\psi_p(x) = \dfrac{x^n - 1}{x - 1} = x^{n-1} + x^{n-2} + \cdots + 1$ for p prime, this defines ψ_n for all n inductively.

Theorem 17. *For any $\theta \in K^\times$ where $K \nmid n$. Let $L : K$ be a splitting field extension for $t^n - \theta$, then:*

- *There is a primitive n^{th} root of unity, ε, in L.*
- *$L/K(\varepsilon)$ is a cyclic extension, and $|\mathrm{Gal}(L/K(\varepsilon))|$ divides n.*
- *$t^n - \theta$ is irreducible over $K(\varepsilon)$ if and only if $|\mathrm{Gal}(L/K(\varepsilon))| = n$.*

Theorem 18. *Let $L : K$ be a cyclic extension of degree n (so $L : K$ is Galois and $G = Gal(L : K) \cong C_n$). Suppose that $\mathrm{char}\, K \nmid n$ and K contains a primitive n^{th} root of unity ε. Then there exists $\theta \in K^\times$ such that $t^n - \theta$ is irreducible over K and $L : K$ is a splitting field extension for $t^n - \theta$.*

The condition $\mathrm{char}\, K \nmid n$ ensures $t^n - \theta$ does not have repeated roots by the derivative test.